Module 2: Nerve & Muscle Physiology. **Imp points**
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TOPIC 1: MUSCLE CONTRACTION (SKELETAL)

. Calcium Storage

- **Question:** Following part of the cell can store calcium to facilitate muscle contraction: (Midterm 2020, 2025)
- **Answer: a) Sarcoplasmic reticulum** (specifically Terminal Cisternae)
  Guyton Explanation: The Sarcoplasmic Reticulum (SR) contains a special protein called *calsequestrin* that allows it to store Calcium in very high concentrations. Release from here triggers contraction.

. Structural Unit (Sarcomere)

- **Question:** The portion of myofibril that lies between two successive Z discs is called a: (Midterm 2020, 2022)
- **Answer: e) Sarcomere**

- **Guyton Explanation:** The Sarcomere is the functional unit of muscle contraction. It is defined as the segment between two Z discs.

. Thick Filaments

- **Question:** Which region of sarcomere contains thick filament only? (Midterm 2022)
- **Answer: c) H band**

- **Guyton Explanation:**
  - **A Band:** Contains Myosin (thick) + overlapping Actin.
  - **I Band:** Contains only Actin (thin).
  - **H Zone:** Central part of A band with **only Myosin** (no actin overlap).

. Contraction Initiation (Calcium Binding)

- **Question:** Binding of calcium to troponin C leads to which change in the muscle fiber? (Midterm 2025, 2022)
- **Answer: C) Exposure of myosin-binding site on actin**

- **Guyton Explanation:** Calcium binds to Troponin C -> Conformational change -> Moves Tropomyosin strand deeper into the groove -> Uncovers the active sites on Actin so Myosin can bind.

. Cross-Bridge Cycle (ATP)

- **Question:** Which step in the cross-bridge cycle requires ATP binding? (Midterm 2025)
- **Answer: D) Detachment of myosin from actin**

- **Guyton Explanation:** The binding of a *new* molecule of ATP to the myosin head causes it to **detach** from the actin filament. (Hydrolysis of that ATP then "cocks" the head for the next stroke).

. Energy Sources

- **Question:** Which energy source is primarily used during the first 10 seconds of maximal muscle contraction? (Midterm 2025)
- **Answer: d) Phosphocreatine system**

- **Guyton Explanation:** The phosphocreatine system provides immediate ATP for 5-10 seconds of maximal power (like a 100m dash). Glycolysis takes over after that.

. Types of Contraction

- **Question:** Which best describes an isometric contraction? (Midterm 2025, 2022)
- **Answer: C) Tension increases without muscle shortening**

- **Guyton Explanation:**
  - **Isometric:** Muscle does not shorten (Same length), but tension increases (e.g., pushing a wall).
  - **Isotonic:** Muscle shortens (Same tone) against a load.

8. Relaxation Mechanism

- **Question:** Which of the following actively contributes to muscle relaxation after contraction? (Midterm 2025)
- **Answer: C) Reuptake of calcium into SR via SERCA**

- **Guyton Explanation:** Relaxation is caused by the active pumping of Calcium *back* into the Sarcoplasmic Reticulum by the SERCA pump (Sarcoplasmic Endoplasmic Reticulum Calcium ATPase).

9. T-Tubule Sensor

- **Question:** Which receptor in the T-tubule senses depolarization in skeletal muscle? (Midterm 2025)
- **Answer: A) DHP receptor (Dihydropyridine)**

- **Guyton Explanation:** The Action Potential travels down the T-tubule and activates the voltage-sensing **DHP receptors**. These mechanically open the **Ryanodine receptors** on the SR to release Calcium.

TOPIC 2: SMOOTH MUSCLE

10. Calcium Mechanism

- **Question:** In smooth muscle fiber Ca++ binds to: (Midterm 2022, 2020)
- **Answer: a) Calmodulin protein**

- **Guyton Explanation:** Smooth muscle has no Troponin. Instead, Calcium binds to **Calmodulin**. This complex activates *Myosin Light Chain Kinase (MLCK)* to phosphorylate myosin and cause contraction.

11. Latch Mechanism

- **Question:** Latch Bridge Mechanism is present in: (Midterm 2020)
- **Answer: e) Smooth muscles**
- **Guyton Explanation:** The "Latch Mechanism" allows smooth muscle to maintain prolonged tonic contraction (like holding a heavy bag or maintaining blood vessel tone) with very little energy expenditure.

12. General Properties

- **Question:** Smooth muscle: (Midterm 2020)

  - Looking at options :

    - a) Rapid contraction (False, it's slow).
    - b) Striated (False).
    - c) Voluntary/Involuntary (False, purely involuntary).
    - d) Found in eye, uterus, blood vessels (True).
    - e) Found in eye and heart (False, heart is cardiac).
    - **Correct Answer: d) Can be found in the eye, uterus and blood vessels**.
    - **Guyton Explanation:** Smooth muscle is non-striated, involuntary, and found in hollow organs (uterus, vessels) and the eye (ciliary muscle/iris).

TOPIC 3: NERVE PHYSIOLOGY

. Refractory Period

- **Question:** Absolute refractory period in muscle/nerve can be defined as a period during which: (Midterm 2020)
- **Answer: a) No impulse can generated A.P**

- **Guyton Explanation:** During the absolute refractory period, Na+ channels are either already open or inactivated. No stimulus, regardless of strength, can trigger a new AP

.

. Action Potential Initiation

- **Question:** First step for initiation of action potential is: (Midterm 2020)
- **Answer: Opening of voltage-gated Sodium channels (Na+ influx)**

. Action Potential Phase 0

- **Question:** Phase (0) occurs due to? (Midterm 2022)
- **Answer: Rapid depolarization (Na+ Influx)**

. Neurotransmitter Release

- **Question:** Which ion is caused for acetylcholine release? (Midterm 2022)
- **Answer: A) Calcium**

- **Guyton Explanation:** Calcium entry into the presynaptic terminal is the specific trigger for vesicle fusion and exocytosis.

. Nerve Fiber Types (Speed)

- **Question:** In nerve fiber classification, fastest conduction occurs in: (Mock 2019)
- **Answer: a) A Alpha**

- **Guyton Explanation:** A-alpha fibers are the largest diameter and heavily myelinated, conducting at 70-120 m/s (used for motor and proprioception). C fibers are slowest (0.5-2 m/s).

. Tetany vs Tetanus

- **Question:** Sustained contractions without relaxation will cause: (Mock 2024, 2023, 2022)
- **Answer: b) Tetanus** (Tetany is usually pathological hypocalcemia spasms; Tetanus is the physiological summation of twitches).

    - *Note:*, "Summation of impulses without any relaxation" is often keyed as **Tetanus** (Physiological) or sometimes **Tetany** (Pathological).

    - **Guyton Definition:** "Tetanization" is when the frequency of stimulation is so high that individual muscle twitches fuse into a smooth, continuous contraction.

. Muscle Fatigue

- **Question:** Sustained contractions without relaxation will cause which of the following conditions? (Mock 2024)
- **Answer: b) Tetanus** (physiological state).

. Treppe (Staircase Effect)

- **Question:** Treppe's phenomenon? (Mentioned in Mock 2024)
- **Guyton Explanation:** When a muscle begins to contract after a long period of rest, its initial strength of contraction increases stepwise (like a staircase). This is due to increasing Calcium ions in the cytosol with each beat.